# Module 2 (Manual Testing)

**1 What is Exploratory Testing?**

**Ans**: It is a type of testing where the tester learns, designs, and executes test cases at the same time without predefined scripts.

**2. What is Traceability matrix?**

**Ans**: Traceability Matrix is a document used to map requirements with test cases to ensure all requirement are tested**.**

**3. What is Boundary value testing?**

**Ans:** Boundary value testing checks the minimum limits of input values

**4. What is Equivalence partition testing?**

**Ans**: Dividing input data into groups and testing one value from each group.

**5. What is Integration testing?**

**Ans:** Integration testing checks whether combined modules work properly together.

**6. What determines the level of risk?**

**Ans:** Risk level is determined by the impact and probability of failure.

**7. What is Alpha testing**

**Ans:** Testing done by the internal team before releasing the software.

**8. What is beta testing?**

**Ans:** Beta Testing is testing performed by real users before final release.

**9. What is component testing?**

**Ans:** Component Testing tests an individual modules or component separately.

**10. What is functional system testing?**

**Ans**: Function system testing checks whether the system works according to functional requirement.

**11. What is Non-Functional Testing?**

**Ans**: Non-Functional Testing is checks performance, usability, and reliability of the system.

**12. What is GUI Testing?**

**Ans**: GUI Testing is checks the graphical interface like buttons, menu, and screen**.**

**13. What is Adhoc testing?**

**Ans**: Adhoc Testing in informal testing done without planning or documentation.

**14. What is load testing?**

**Ans**: Load Testing is checks the system performance under expected user load.

**15. What is Stress testing?**

**Ans:** Stress testing checks system behavior under expected user load.

**16. What is white box testing and list the types of white box testing?**

**Ans**: White Box Testing is internal code and logic of the software. Types: Statement coverage, Decision coverage, Condition Coverage

**17. What is black box testing? What are the different black box testing techniques?**

**Ans:** Black Box testing is software without knowing internal code, focusing on input and outputs.
Different black box testing techniques:
Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing,
State Transition Testing

**18. Mention what are the categories of defects?**

**Ans**: Categories of defects are Functional, Performance, Usability, Security, and Compatibility defects.

**19. Mention what big bang testing is?**

**Ans**: Big Bang testing combines all modules together and tests them at once.

**20. What is the purpose of exit criteria?**

**Ans**: It is used to decide whether testing is completed or not.

**21. When should "Regression Testing" be performed?**

**Ans:** Regression testing is performed after changes or bug fixes in the application.

**22. What is 7 key principles? Explain in detail?**

**Ans**: 1. Testing shows presence of defects- Testing finds defects but cannot prove software is defect-free.

2. Exhaustive testing is impossible- It is not possible to test everything.

3. Early Testing- testing should start as early as possible.

4. Defect clustering- Most defects are found in a few modules.

5. Pesticide paradox- Repeating the same tests finds fewer new defects.

6. Testing is context dependent-Testing approach varies by application.

7. Absence of error fallacy- Defect-free software is useless if it does not meet user needs.

**23. Difference between QA v/s QC v/s Tester**

**Ans:**

| Aspect | QA(Quality Assurance) | QC(Quality Control) | Tester |
|---|---|---|---|
| Focus | Process-oriented | Product-oriented | Testing activities |
| Goal | Prevent defects | Find defects | Execute test cases |
| Example | Setting standards & processes | Reviewing product | Finding bugs |

**24. Difference between Smoke and Sanity?**

**Ans:**

| Aspect | Smoke Testing | Sanity Testing |
|---|---|---|
| purpose | Check basic build stability | Verify specific changes/bug fixes |
| Coverage | Broad and shallow | Narrow and deep |
| Goal | Ensure system is testable | Ensure fixes are working |

**25. Difference between Verification and Validation?**

**Ans:**

| Aspect | Verification | Validation |
|---|---|---|
| Meaning | Checks process & documents | Checks actual product |
| Types | Are we building it right? | Are we building the right product? |
| Example | Reviews, inspection | Execution test cases |

**26. Explain types of Performance testing?**

**Ans:** Types of Performance Testing are Load, Stress, Spike, Endurance, and Volume testing.

**27. What is Error, Defects, Bug and failure?**

**Ans:** Error: Human mistake, Defect/Bug: Problem in software, Failure: Software unable to perform expected result.

**28.. Difference between Priority and Severity?**

**Ans:**

| Aspect | Severity | Priority |
|---|---|---|
| Meaning | Impact of defect on system | Urgency to fix defect |
| Focus | Technical issue | Business need |
| Example | System crash= High severity | Minor UI issue but urgent = High priority |

**29. What is Bug Life Cycle?**

Ans: Bug life cycle is the process through which a bug goes from identification to closure.
Stages: New -> Assigned -> Open -> Fixed -> Retest -> Closed

**30. Explain the difference between Function testing and Non-Function testing?**

**Ans:**

| Aspect | Functional Testing | Non-Functional Testing |
|---|---|---|
| Focus | Features & functions | Performance & behavior |
| Checks | What system does | How system works |

**31. What is the difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)?**

**Ans:**

| Aspect | STLC | SDLC |
|---|---|---|
| Focus | Testing process | Development process |
| Performance by | Testers | Developers and Testers |
| Goal | Find defects | Develop software |

**32. What is the difference between test scenario, test cases, test script?**

**Ans:**

| Aspect | Test Scenario | Test Case | Test Script |
|---|---|---|---|
| Meaning | High-level testing idea | Detailed testing steps | Automated test instructions |
| Focus | What to test | How to test | Execute testing automatically |
| Example | Check Login featurs | Enter valid username & password | Automation script for login |

**33. Explain what Test Plan? What is the information that should be covered.**

**Ans**: A Test plan is a document that describes the testing objectives, scope strategy, and schedule of a project.

**34. What is priority?**

**Ans**: Priority shows how urgently a defect should be fixed.

**35. What is Severity?**

**Ans:** Severity shows the impact of a defect on the application.

**36. What is Bug categories are?**

**Ans:** Bug Categories: Functional, Performance, UI, Security, and Compatibility bugs.

**37. Advantages of Bugzilla?**

**Ans:** Easy bug tracking, Generates reports, Email notification.

**38. Difference between priority and severity**

**Ans:**

| Aspect | Severity | Priority |
|---|---|---|
| Meaning | Seriousness of defect | Urgency to fix defect |
| Focus | System impact | Business need |
| Decided by | Tester | Manager/Client |

**39. What are the different Methodologies in Agile Development Model?**

**Ans**: 1 Scrum 2.Kanban 3.Extreme Programming 4.Lean Development 5. Crystal Method 6.Feature Driven Development 7. Dynamic System Development Method

**40. Explain the difference between Authorization and Authentication in Web testing. What are the common problems faced in Web testing?**

**Ans:**

| Aspect | Authentication | Authorization |
|---|---|---|
| Meaning | Verifies user identify | Verifies user permissions |
| Checks | Username and Password | Access rights |
| Example | Login process | Access to admin page |